Below Zero: Supplementary Material

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This supplement provides implementation details, regional analyses, calibration, and development-set controls. Numbering continues from the main text; full-precision scores and source-image identifiers are included with the accompanying data.

I. Units, Conditioning, and the Meaning of a Sample

A. Signal Versus Readout

The network output $s \geq 0$ is an expected signal in DN; the corresponding electron mean is s/g. With $N \sim \text{Poisson}(s/g)$, the simulator's integer path is

$$\begin{aligned} A &= \min\{16383, \lfloor 512 + gN + \epsilon + b + 1/2 \rfloor\}, \\ V &= A - 512. \end{aligned} \tag{14}$$

All admitted simulated samples satisfy the physical lower ADC bound. Four-phase offsets are independently uniform over $[-0.25, 0.25]$ DN and enter the estimator and decoder as known conditions.

A zero observation implies $V_i \leq 0$. Under independent sensor noise, neighboring observations inform the latent signal; the conditional law at the estimated signal supplies the missing readout distribution.

Let $m = \mathbb{E}[V \mid Y, s]$ and $v = \text{Var}(V \mid Y, s)$ at a censored pixel. For an independent draw V' from the same conditional law,

$$\mathbb{E}[(V - m)^2 \mid Y, s] = v, \tag{15}$$

$$\mathbb{E}[(V - V')^2 \mid Y, s] = 2v. \tag{16}$$

A squared-error criterion therefore favors the conditional mean, which suppresses variation, over a statistically correct draw. This motivates the distribution scores used in the main comparison.

B. Excess Scores and the Known-Signal Reference

For the known-signal conditional law $p_i^\star$ and a prediction q_i on a common discrete support,

$$\mathbb{E}_{V \sim p_i^\star}[-\log q_i(V)] - H(p_i^\star) = D_{\text{KL}}(p_i^\star \| q_i). \tag{17}$$

For the ranked probability score, expansion of the squares gives

$$\begin{aligned} &\mathbb{E}_{V \sim p_i^\star}[\text{CRPS}(q_i, V) - \text{CRPS}(p_i^\star, V)] \\ &\quad = \Delta \sum_k \left(F_{q_i}(k) - F_{p_i^\star}(k) \right)^2 . \end{aligned} \tag{18}$$

These identities separate conditional uncertainty from distribution-estimation error. The known-signal law has zero self-divergence and cumulative discrepancy, while its realized NLL remains nonzero.

For a negative label, zero fill has ranked score $|V|$ and infinite NLL. The main table reports these values directly.

TABLE V
N0 architecture in packed CFA coordinates.

Stage	Width	Blocks	Spatial scale
Input stem	32	—	1
Encoder 1	32	2	1
Encoder 2	64	2	1/2
Encoder 3	128	4	1/4
Encoder 4	256	4	1/8
Bottleneck	512	4	1/16
Decoder 1–4	256/128/64/32	2 each	1/8 to 1
Signal head	4	—	1

Inputs: 4 transformed measurements, 4 zero indicators, 4 offset maps, 1 validity mask, and 3 constant maps. Output: 4 softplus signal maps. The 10,913,380-parameter model uses no pooled channel or spatial attention.

II. Network, Data, and Training Details

A. Network Architecture

We pack the Bayer phases as $[R, G_1, G_2, B]$ and provide 16 input channels: four $\text{asinh}(Y/4)$ measurements, four zero-candidate indicators, four broadcast offset maps, a validity mask, and constant maps for g, σ_r, and a reserved zero. The validity mask distinguishes padded pixels from measured zeros. The known offsets enter both the estimator and the PG decoder.

Table V gives the NAF-style architecture. Each normalization operates across channels at one spatial position, using FP32 arithmetic and $\epsilon = 10^{-6}$. SimpleGate multiplies two channel halves, and learned residual scales are initialized to zero.

Nested 128/256 contexts share a 64-pixel target, even Bayer alignment, and a padding-validity mask. Flips and rotations precede packing.

B. Synthetic Distribution and Split Integrity

For the calibrated profile, the measured integer-domain read-noise variance is corrected by $1/12\,\text{DN}^2$ before the explicit 1-DN quantizer is applied in simulation.

The 1,100 rendered-RGB templates split into 825/137/138 training/development/test groups. Natural draws (90%) use median-normalized log-uniform exposure over 0.4–64 DN and color factors $\exp U[-0.3, 0.3]$; 30% add a 0.1–2 DN floor and 5% are zero-light controls. Procedural draws supply 10%. These settings are independent of the real exposures.

Each source has two cases with different crop, augmentation, exposure, and noise seeds. The methods share each case, and the contexts share its center; bootstrap resampling keeps all four related conditions together.

C. Optimization and Precision

AdamW uses a 2×10^{-4} peak learning rate, 2,000-update warmup, cosine decay to 2×10^{-6} at 80,000 updates, weight decay 0.01, gradient clipping at 1, and moving-average decay 0.999. Two physical batches of size one are accumulated per update, each contributing at most 2,048 zero candidates to the conditional cross-entropy loss. The selected moving-average checkpoint is update 80,000 of seed 7103.

The physical decoder evaluates 306 category probabilities from the four predicted signal maps. Training uses mixed precision; final network inference uses FP32 with TF32 disabled. Likelihood and mixture routines use FP64 where required.

D. Score Aggregation and Confidence Intervals

We average candidate scores within each case, nonempty cases within each source, and then sources with equal weight. Paired bootstrap resampling treats a source and all its cases as one unit. Aggregate comparisons use 10,000 resamples. The preregistered KL contrasts against Local Bayes and Foi-qPG use 97.5% percentile intervals, controlling the familywise error rate at 5% across the two comparisons; secondary comparisons use 95% intervals. Regional comparisons are exploratory and use the pointwise 95% procedure in Section IV-B. Development and regional intervals have no multiplicity adjustment. All intervals condition on the trained model and quantify source-image sampling variability.

III. Tail Resolution and Probability Calibration

The lookup grid uses steps of 0.002 in $\log(1 + s)$ and 0.01 DN in b. Across the 152 archived validation queries, the maximum teacher-to-interpolant divergence was 2.18×10^{-7} nats and the maximum mass-normalization error was 7.77×10^{-16}.

TABLE VI
Discrete-interval calibration on 32 development sources. Predicted and observed values are percentages. Difference is observed coverage minus actual predicted interval mass, in percentage points.

Method	Nominal	Predicted	Observed	Difference
N0/PG	50%	52.584	52.460	−0.123
N0/PG	80%	81.864	81.510	−0.354
N0/PG	90%	91.819	91.685	−0.134
N0/PG	95%	97.632	97.597	−0.035
Local PG	50%	52.177	51.962	−0.215
Local PG	80%	81.981	81.509	−0.472
Local PG	90%	91.929	91.658	−0.270
Local PG	95%	97.639	97.538	−0.101
Known signal	50%	52.496	52.378	−0.118
Known signal	80%	81.864	81.513	−0.352
Known signal	90%	91.854	91.725	−0.129
Known signal	95%	97.635	97.601	−0.034

A. Resolving Codes Within the Tail

Let $K = 304$ and $T = \{V < -K\}$. The stored law is

$$\bar{q} = (q(0), q(-1), \ldots, q(-K), q(T)). \tag{19}$$

Excess KL compares the normalized coarsened laws $\bar{p}^{\star}$ and $\bar{q}$. It includes the tail mass but does not distinguish different distributions within T.

If category T is selected, a second independent coordinate substream draws an integer from the PG law conditional on T. For a withheld tail code $k < -K$, the log probability is

$$\log q(k) = \log q(T) + \log p_{\mathrm{PG}}(k) - \log P_{\mathrm{PG}}(T). \tag{20}$$

For Bayesian mixtures, this calculation retains every signal component's tail law and posterior weight. Substitution of a mean signal would give a different distribution.

Residual bounds for the admitted moment and ranked-score calculations are below 10^{-12}. A -400 DN test compares mixture-tail evaluation with independent scalar laws. Real tail labels are likewise evaluated at their actual codes.

B. Calibration Versus Sampler Verification

Table VI reports coverage on the fixed 32-source development cohort. For discrete equal-tail intervals, empirical coverage is compared with the interval's actual predicted mass.

Computing the sample variance at a given location requires multiple draws. With fixed laws and coordinate keys, reversing order or changing to chunks of size 257 leaves the sampled output unchanged. All outputs preserve positive codes, fill candidates with nonpositive integers, and reproduce the input after clipping.

We test 2,048 fixed predicted laws with two random streams and 1, 8, 32, and 64 draws; the reference RMSE is $\sqrt{\mathrm{Var}/S}$. Sample-mean RMSE falls from 15.164 DN at one draw to 1.949 DN at 64 draws, close to

the expected $S^{-1/2}$ scaling. At 64 draws, the mean of the per-location sample variances is 227.397 DN^2, compared with 226.970 DN^2 predicted. The sampling curve includes both random streams.

IV. Input Information and Additional Spatial Analyses

A. Local Likelihood and Bayesian Integration

For each same-CFA neighborhood $\mathcal{N}_i$, let h_{it} be the count of observed code t. The local signal likelihood is

$$\ell_i(s) = h_{i0} \log c(s, b) + \sum_{t>0} h_{it} \log p_Y(t \mid s, b). \quad (21)$$

Upper saturation contributes an event mass to the likelihood. Local PG refines a grid maximum; Local Bayes integrates conditional laws using the training-fitted zero-plus-Gamma signal prior. The center event enters once. The signal grid uses approximately 0.1 DN spacing below 32 DN and 0.5 DN above, with conversion to electron means by g.

N0 uses four-phase multiscale context; Local/Foi use same-phase observations. Joint-CFA Bayes (Section VII) uses observations from all four CFA phases with a 5×5 model and a finite empirical prior.

B. Held-Out Regional Definitions

Regional analysis of the held-out test uses same-CFA 9×9 neighborhoods. With $u = \log(1 + s)$, signal variation is $t = \sqrt{\overline{u^2} - \bar{u}^2}$, normalized over valid pixels. Thresholded gradient locations have Sobel magnitude at least 0.15 after division by eight and a one-pixel validity erosion; distance is Euclidean in packed pixels. The observed zero fraction is the valid-neighborhood mean of $\mathbf{1}[Y = 0]$.

Signal-bin edges in DN are $0, .25, .5, 1, 2, 4, 8, 16, \infty$; variation edges are $0, .025, .05, .1, .2, .4, .8, \infty$; distance edges are $0, 1, 2, 4, 8, 16, \infty$; zero-fraction edges are $0, .1, .25, .5, .75, .9, 1$. The 256-pixel context supplies 1,627,306 unique candidates. For each source and bin, we divide the score sum across both cases by their combined candidate count, then average the available source means equally. Pointwise 95% intervals use 2,000 paired source resamples (seed 1280027). Source membership therefore varies by bin. The $z \geq 0.75$ bin has two candidates from one source; its scores and all other bin counts are included in `V9_heldout_spatial_bins_256.csv`.

The boundary case $s < 0.25$ DN contains 311,545 candidates from 94 sources. N0 minus Foi-qPG KL is 5.35×10^{-5} nats, with interval $[2.68, 9.88] \times 10^{-5}$. The corresponding Local Bayes contrast is -8.72×10^{-5}, with interval $[-25.0, 0.693] \times 10^{-5}$. Near-black performance therefore differs markedly from the gains observed in structured regions.

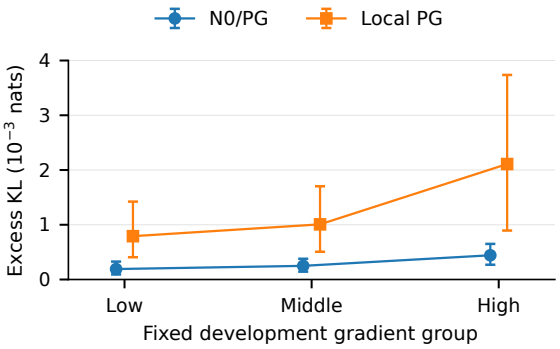


Fig. 5. Absolute conditional-distribution errors on the fixed 32-source development set. Points and 95% source-bootstrap intervals use equal source weighting within each gradient group. Both errors rise with gradient; Local PG rises more. The cohort and grouping differ from the held-out marginal comparisons in Fig. 4 of the main text.

C. Development Absolute Errors and Matched Contrasts

Fig. 5 reports absolute source-mean KL for the fixed low, middle, and high gradient groups. N0/Local PG values are $0.000191/0.000791$, $0.000249/0.001007$, and $0.000442/0.002109$ nats. The mean change in the paired KL difference (N0 minus Local PG) from the low- to the high-gradient group is -0.001066, with median -0.000102 and 23/32 negative source values; the two largest reductions contribute 65.7% of the net sum. Its 10,000-resample source interval is $[-0.002177, -0.000271]$. The NLL contrast is -0.000807, with interval $[-0.003231, 0.001221]$ and 12/32 negative values.

Coarse signal matching retains source–CFA–signal-bin cells with at least five low- and five high-gradient candidates. Cells receive the smaller count as weight within each source; sources are averaged equally. Across 142 cells from 27 sources, KL has mean -0.000268, median -0.0000358, and 18/27 negative values; its 20,000-resample source interval is $[-0.000581, -0.0000280]$. NLL is 0.000122, interval $[-0.002049, 0.002086]$. The accompanying data give every matching variant and source effect; Table VII reports regional contrasts that favor a comparator or have intervals crossing zero.

D. Why Pooled Curves Can Overlap

For equally weighted distributions,

$$\mathrm{TV}\left(\frac{1}{n}\sum_i q_i, \frac{1}{n}\sum_i r_i\right) \leq \frac{1}{n}\sum_i \mathrm{TV}(q_i, r_i). \quad (22)$$

The triangle inequality gives this bound. Development mean pointwise TV is 0.01235; real pooled-PMF TVs are 0.000282 and 0.001561 on different cohorts and weights. The archived pointwise histogram retains the full diagnostic distribution. Paired scores evaluate the accuracy of the locationwise predictions.

TABLE VII

SELECTED REGIONAL NLL CONTRASTS THAT FAVOR A COMPARATOR OR HAVE INTERVALS CROSSING ZERO. N0-MINUS-COMPARATOR DIFFERENCES AND POINTWISE 95% SOURCE-BOOTSTRAP INTERVALS ARE IN 10^{-3} NATS. THE FIRST TWO ROWS ARE POST-HOC HELD-OUT CASE GROUPS WITH ZERO FRACTION $z_{\text{case}} \geq 0.5$; THE REMAINING ROWS USE DEVELOPMENT CANDIDATES. HERE d IS THE SOBEL MAGNITUDE OF $\log(1+s)$ DIVIDED BY EIGHT, AND a IS THE FOUR-CONNECTED ZERO-COMPONENT SIZE IN PACKED PIXELS.

Feature / bin	Comparator	Sources	Difference	95% interval	
Case zero fraction ≥ 0.5	Local Bayes	30	$+0.0020$	-0.0132	$+0.0172$
Case zero fraction ≥ 0.5	Foi-qPG	30	$+0.0080$	-0.0039	$+0.0200$
Signal $0 \leq s < 2$ DN	N0 permuted	28	$+1.3432$	-2.3007	$+6.1834$
Signal $s \geq 16$ DN	Local PG	20	-1.9514	-4.0343	$+0.2155$
Gradient $0 \leq d < 0.025$	Local PG	32	$+0.0043$	-1.4801	$+1.6993$
Gradient $0.025 \leq d < 0.1$	Local PG	32	$+0.1099$	-0.9157	$+1.2081$
Gradient $d \geq 0.1$	Local PG	32	-0.8031	-2.0314	$+0.3147$
Zero-component size $a \geq 33$	Local PG	30	-0.5950	-2.0038	$+0.5810$
Zero-component size $a \geq 33$	N0 permuted	30	-0.5265	-3.8384	$+3.0659$

TABLE VIII

REAL-DATA NLL AND CRPS DIFFERENCES UNDER TAIL AND REPEATED-COORDINATE EXCLUSIONS. N0 MINUS LOCAL PG ON COMMON R/G_1/G_2 CANDIDATES; NEGATIVE VALUES FAVOR N0. THE ALL-EVENT ROW GIVES THE FULL-SAMPLE RESULT, FOLLOWED BY SUBSETS SELECTED USING THE RETAINED READOUTS.

Frame	Included positions	N	$10^3\Delta$NLL	ΔCRPS (DN)
4748	All	12,652	-1.342	-0.01102
	$V > -128$	12,609	-1.375	-0.01145
	Excl. repeat coords.	12,650	-1.336	-0.01098
4749	All	19,200	-0.964	-0.00920
	$V > -128$	19,121	-0.998	-0.00927
	Excl. repeat coords.	19,198	-0.945	-0.00907

TABLE IX

ZERO-ATOM AND TAIL EVENTS AMONG ALL 31,852 SHARED REAL R/G_1/G_2 CANDIDATES. EXPECTED COUNTS SUM N0 PROBABILITIES; OBSERVED COUNTS USE THE RETAINED SIGNED LABELS.

Event	Expected count	Observed count
$V = 0$	1107.76	1190
$V \leq -1$	30744.24	30662
$V \leq -50$	1345.75	1065
$V \leq -125$	0.01690	132
$V \leq -305$	8.63×10^{-30}	13

V. REAL-EXPOSURE AND DOWNSTREAM PROTOCOLS

A. Integer Alignment and Hidden References

Real references use the fixed coordinate transformation $V = R_{\text{native}} - 512 + n_c$, with $n_c = (2, 3, 3, 3)$ for R/G_1/G_2/B. The original camera files remain unchanged; integer shifts preserve the quantization grid.

Independent dark-frame calibration leaves fractional residuals $(-0.437, -0.310, -0.189, +0.078)$ DN. The network and decoder retain zero inference offsets, so these residuals remain part of the real-data model mismatch.

Table IX reports the fixed -125 DN diagnostic, distinct from the main-text -128 threshold. The events are nested. Section VII-D relates the extreme labels to their original decoded codes.

B. Fixed Demosaicking Pipeline

The fixed LibRaw 0.22.1 pipeline calls `open_bayer(black=0)`, `unpack`, `raw2image_ex(0)`, `pre_interpolate`, and `ahd_interpolate`, followed by FP32 pedestal subtraction. This bypasses ordinary black subtraction, scaling, white balance, color/gamma conversion, and denoising. Revision and patch hashes are recorded with the renderer.

Negative-pattern tests verify that below-black information reaches interpolation. In the recorded renderer validation, the fixed-source implementation reproduced all six full-frame linear RGB arrays exactly.

AHD's nonlinear decisions depend on the pedestal, as verified at $P = 512$ and $P = 1024$. All comparisons use $H(V + 512) - 512$.

C. A/B/C/D Statistics

Each real ROI is 512×512 Bayer pixels. Removing a 16-pixel border leaves 480×480 evaluated RGB pixels. A and B are fixed signed/clipped references. Each of the 32 C/D draws is demosaicked separately; spatial statistics are computed per draw before aggregation. D uses the real-data Local PG 3×3 window; the synthetic test uses 5×5.

Table X reports the two preselected darkest ROIs. Each draw contributes its own spatial mean and standard deviation; intervals describe variation across draws. Spatial standard deviations include scene texture and demosaicking correlation.

D. Processing-Order Illustration

Supplementary Visual A shows the P2654749 crop at $(0, 1500)$: 1000×1000 Bayer pixels, full-frame AHD, and fourfold area reduction to 250×250 pixels. The labeled routes differ in RAW/RGB clipping order and share exposure and display transforms.

TABLE X
LINEAR CAMERA-RGB STATISTICS IN THE TWO PRESELECTED DARKEST REAL ROIS, BEFORE WHITE BALANCE OR DISPLAY CLIPPING. A/B ARE FIXED REFERENCES; C/D AVERAGE THE SPATIAL STATISTICS OF 32 SEPARATELY DEMOSAICKED SAMPLES. UNITS ARE DN. THE DATA CONTAIN ALL 12 ROIS.

Exposure	AHD input	μ_R	μ_G	μ_B	sd$(R-G)$	sd$(B-G)$
P2654748	A: signed reference	8.041	9.003	3.200	25.986	22.037
	B: clipped	15.368	16.322	11.707	18.635	15.127
	C: N0 samples	7.924	8.621	1.655	25.157	22.495
	D: Local PG samples	7.933	8.602	1.642	25.256	22.587
P2654749	A: signed reference	0.423	0.797	-0.225	23.584	20.529
	B: clipped	10.089	10.287	9.146	15.243	12.712
	C: N0 samples	0.231	0.246	-1.871	22.813	20.860
	D: Local PG samples	0.352	0.352	-1.812	22.799	20.861

E. Fixed Clean-Reference Color Endpoint

This development experiment followed the earlier analyses, with its protocol fixed before the color scores were computed. The protocol specifies the renderer and error metrics, with no preset perceptual-effect threshold. Eight geometrically complete natural crops were selected from the 32-source development set without using endpoint scores; four procedural templates were evaluated separately. Each scene generated 16 independent PG sensor realizations and two conditional draws per estimator and realization. The renderer used integer AHD with pedestal 512, fourfold signed-domain integration, fixed white balance and color matrix, and one clean-reference exposure per scene. After excluding the score border, the eight blocks with lowest clean-reference luminance formed the evaluation region. We measured squared block-mean $(R-G, B-G)$ error in exposure-scaled linear display RGB against clean-Q-AHD, where $Q(s) = \lfloor s + 0.5 \rfloor$. Conditional means were quantized with the same Q before AHD. We scored each draw separately, averaged within sensor realization, and then weighted the eight natural sources equally.

N0 sampling reduces dark-color error by 0.003139 relative to Local PG (paired source-bootstrap 95% interval $[-0.006136, -0.000843]$), with lower error on all eight natural sources. The two largest reductions contribute 67.9% of the total improvement. Quantized-mean completion reduces the N0 color error by a further 16.5% relative to sampling. For clean-RAW MSE, N0 sampling instead exceeds Local PG by 2.062 DN^2 (95% source interval $[0.713, 3.834]$); only one of eight source differences favors N0. The signed arm uses the uncensored readout of each of the 16 sensor realizations, with clean-Q-AHD as a separate color reference. Its paired color-error difference from N0 spans zero. All 12 sources, secondary scores, and per-draw records are supplied; the eight-source values are also collected in `evidence/clean_color/PER_SOURCE.csv`.

VI. NUMERICAL AND REPRODUCTION RECORDS

Inference uses a 256-pixel core and a 192-pixel halo in packed CFA coordinates. Core and halo sizes are multiples of 16, preserving the encoder phase. Measurements and validity masks are zero-padded to a multiple of 16; tiles are clipped at the image boundary, and only their core predictions are retained. The released implementation specifies this procedure in `inference/tiling.py`. Sampling uses absolute pixel coordinates as described in Section III. The accompanying records retain decoder identities and reproduction tolerances.

TABLE XI
POST-HOC DEVELOPMENT COMPARISON ON 32 FIXED SOURCES. CFA MEANS ARE COMPUTED WITHIN SOURCE, FOLLOWED BY EQUAL SOURCE WEIGHTING.

Method	Excess KL (nats)	NLL (nats)	CRPS (DN)
N0/PG	0.000376673	3.911227506	8.103882756
Joint-CFA local Bayes	0.001790969	3.912282251	8.112198163
Same-CFA Local PG	0.001952344	3.912362493	8.112870272

VII. ADDITIONAL DEVELOPMENT CONTROLS

The following post-hoc controls use development data and supplement the registered independent test.

A. Joint-CFA Local Development Comparator

Joint-CFA local Bayes uses 256 equally weighted four-phase signal templates from 128 training scenes, with two templates per scene. A packed 5×5 likelihood includes all valid four-phase observations and counts each censoring event once. Prediction mixes the corresponding conditional integer PG laws. N0 and both local estimators are evaluated at the same 32,462 positions from 32 development sources.

N0 minus Joint-CFA differences, with 95% source-bootstrap intervals, are: KL -0.00141430 $[-0.00296235, -0.00031761]$; NLL -0.00105475 $[-0.00215630, -0.00029750]$; and CRPS -0.00831541 DN $[-0.01755261, -0.00217521]$. Joint-CFA minus Local PG NLL and CRPS intervals cross zero.

B. Runtime and Resource Table

Timings use an RTX 3090 Ti, Python 3.10, PyTorch 2.6/CUDA 12.4, two CPU threads, and synchronized

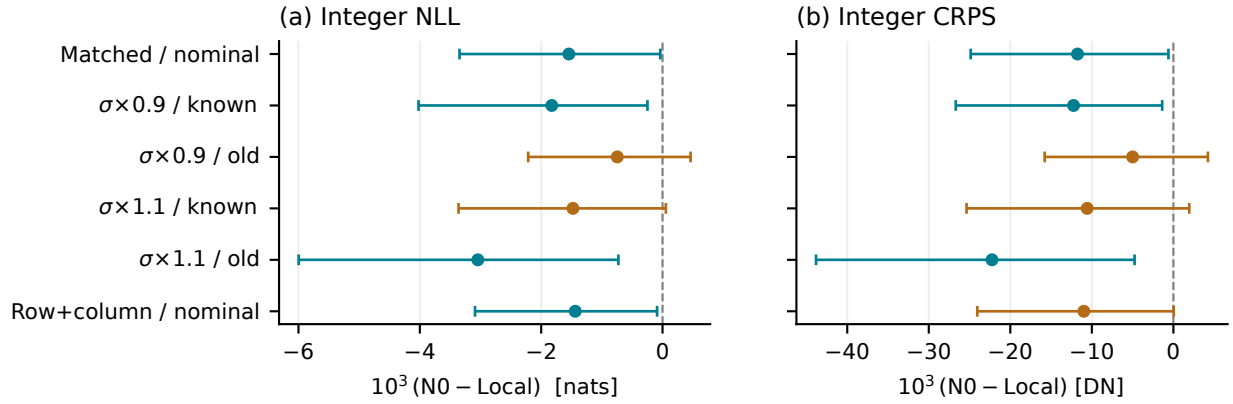


Fig. 6. Paired calibration and correlation tests on 32 development sources. Each setting uses 8,192 central zero candidates shared by both methods. Points show mean differences averaged first over CFA phases and then over sources, with 95% source-bootstrap intervals; negative values favor N0. "Known" supplies the changed profile to both estimators and their decoders; "old" retains the original calibration. Row/column offsets are unobserved. Orange intervals cross zero.

TABLE XII
IN-MEMORY IMPLEMENTATION COST. DECODING, INITIALIZATION, AND FILE OUTPUT ARE EXCLUDED; BACKENDS AND WARM-UP DIFFER AS SPECIFIED IN THE TEXT.

Metric	N0/PG	Local PG 3×3
128 packed median (s)	0.2074	0.7329
24MP median of 3 (s)	27.3901	205.1503
24MP range (s)	25.1087–27.9992	194.3892–219.9394
CUDA allocated (MiB)	750.69	94.84
CUDA reserved (MiB)	1112.00	254.00

GPU stages. The 24MP input has 2,637,992 zeros. Patches use 10 warmups/30 runs; N0 full-frame uses 10 warmups, while Local PG combines one pilot/two later runs. N0 uses FP32 network/GPU LUT; Local uses FP64 GPU likelihood/CPU LUT; both sample on CPU. Full timings are archived.

Local PG core/halo is 128/1. At 32 checked positions, signal error is below 3.05×10^{-12} DN and LUT PMFs match. The checked tile MLEs remain at least 179.14 DN below the grid limits.

C. Paired Calibration and Correlation Probes

N0 and same-CFA 5×5 Local PG use the same generated inputs from 32 development sources: a 128-pixel packed context, central 64-pixel region, and at most 64 zeros per phase. Scores are averaged over CFA phases and then sources. The paired 95% intervals use 10,000 source resamples. Candidates are shared between methods within each setting and can differ between settings.

At read-noise factors 0.9/1.1, both methods receive either the true new profile or the old profile; known-profile network constants extrapolate beyond training. Gaussian row/column offsets each have 5 DN standard deviation and are withheld from both methods. That condition reports no oracle-KL. Integer scoring resolves the PG tail.

Fig. 6 reports all conditions. Old calibration at 0.9σ gives NLL -0.000746 nats, interval $[-0.002214, 0.000462]$; known 1.1σ also has inconclusive contrasts. Row/column offsets give NLL -0.001439, $[-0.003090, -0.000090]$, while the CRPS interval crosses zero. The accompanying data include full scores, profile probes, and rare-tail tests.

D. Real-Code Event Scope

All 31,852 primary real candidates remain scored. The nested $-128/-256/-305$ thresholds contain 122/21/13 events. The deepest 13 decode to native 79–202 and aligned -430 to -307 DN. Two coordinates repeat below -128, one below -305; the full frames separately contain 18/25 native zeros. The ten dark frames supply calibration. Although the deepest events contribute 0.03917/0.04525 nats to absolute N0 NLL, they change the paired difference by only $+0.0000040/-0.0000083$. Removing $V \leq -128$ leaves NLL differences of $-0.001375/-0.000998$ (Supplementary Table VIII).

N0 has lower NLL in 7/12 fixed ROIs. Symmetric 1% winsorization gives differences of $-0.001269/-0.001082$, compared with $-0.001342/-0.000964$ for all events. B-phase diagnostic N0/Local NLL is 3.807768/3.810883 and 3.857925/3.859954 in the two exposures, using the shared R/G_1 noise calibration. The data retain all ROI/CFA contributions.